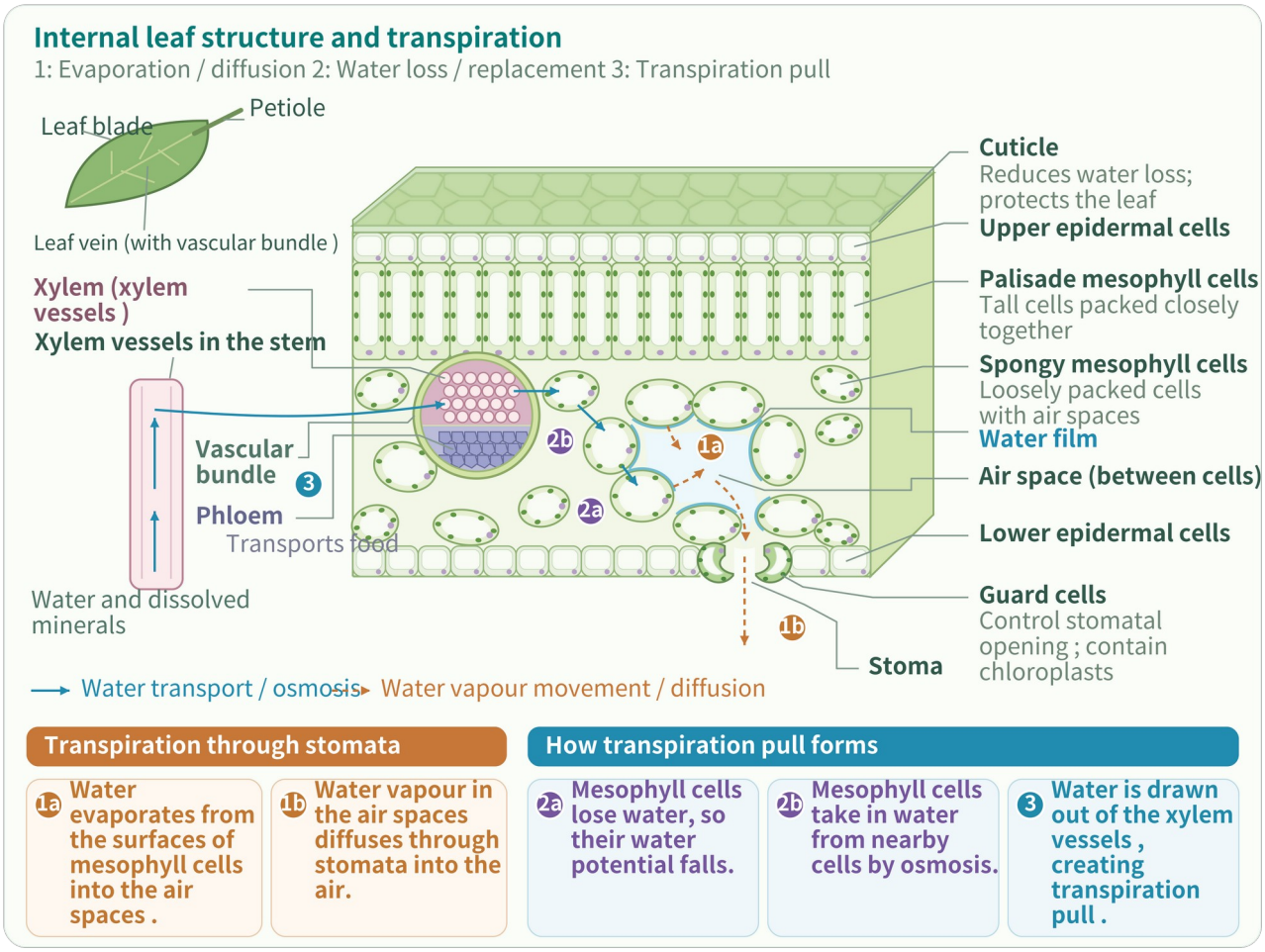
Worksheet 2: Transpiration and water transport

Name: _____ Class: _____ No.: _____

Topic notes



1. Transpiration: water films evaporate; vapour diffuses through stomata.

2. Water replacement: mesophyll water potential falls; osmosis draws in water.

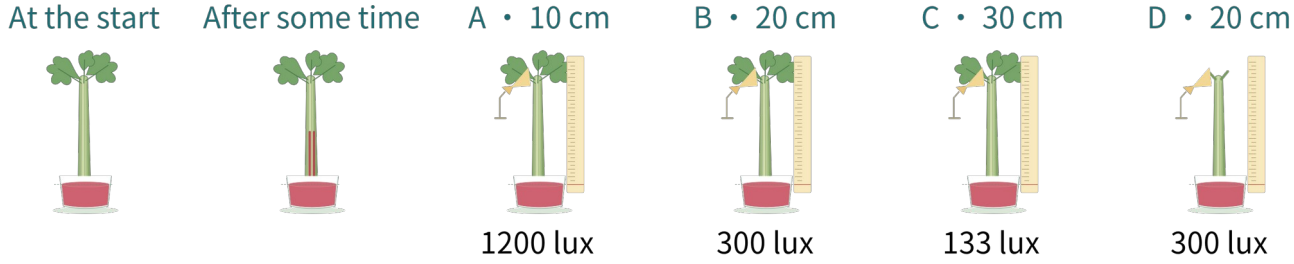
3. Upward transport: removing xylem water creates transpiration pull.

With adequate water and other conditions equal, stronger light favours **stomatal opening**.

A celery stalk is a **petiole**.

Part 1 | Context and fair comparison

The gardening club asks: do light and leaves affect upward water transport?



1. Identify the variables

Tick the appropriate factors in each row. You may choose more than one per type.

Variable type	Factors
Independent variable the factor changed on purpose	☑ **Light intensity (光強度)** ☐ Height / average upward speed of the red water trace ☐ Temperature ☐ Humidity (濕度) and air movement ☐ Red water concentration and depth of immersion (浸入深度) ☐ Total leaf area of the setups with leaves ☐ Length and thickness of the celery stalks
Dependent variable the result measured	☐ Light intensity (光強度) ☑ **Height / average upward speed of the red water trace** ☐ Temperature ☐ Humidity (濕度) and air movement ☐ Red water concentration and depth of immersion (浸入深度) ☐ Total leaf area of the setups with leaves ☐ Length and thickness of the celery stalks
Controlled variables the factors kept the same	☐ Light intensity (光強度) ☐ Height / average upward speed of the red water trace ☑ **Temperature** ☑ **Humidity (濕度) and air movement** ☑ **Red water concentration and depth of immersion (浸入深度)** ☑ **Total leaf area of the setups with leaves** ☑ **Length and thickness of the celery stalks**

2. Select the assumptions for a fair comparison (select all that apply)

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

☑ **Changing the lamp distance does not make the setups have different temperatures.**

☑ **Setups A, B and C have the same total leaf area.**

☐ Leaf area does not affect the results, so it does not need to be kept the same.

☑ **Humidity (濕度) and air movement are the same for all setups.**

3. A comparison setup

How would you plan a comparison setup to find out whether leaves help with water transport?

(Hint: the control group provides a basis for comparison. It helps check whether the change in the experiment is caused by the independent variable rather than other external factors. The control group therefore does not receive the treatment being tested.)

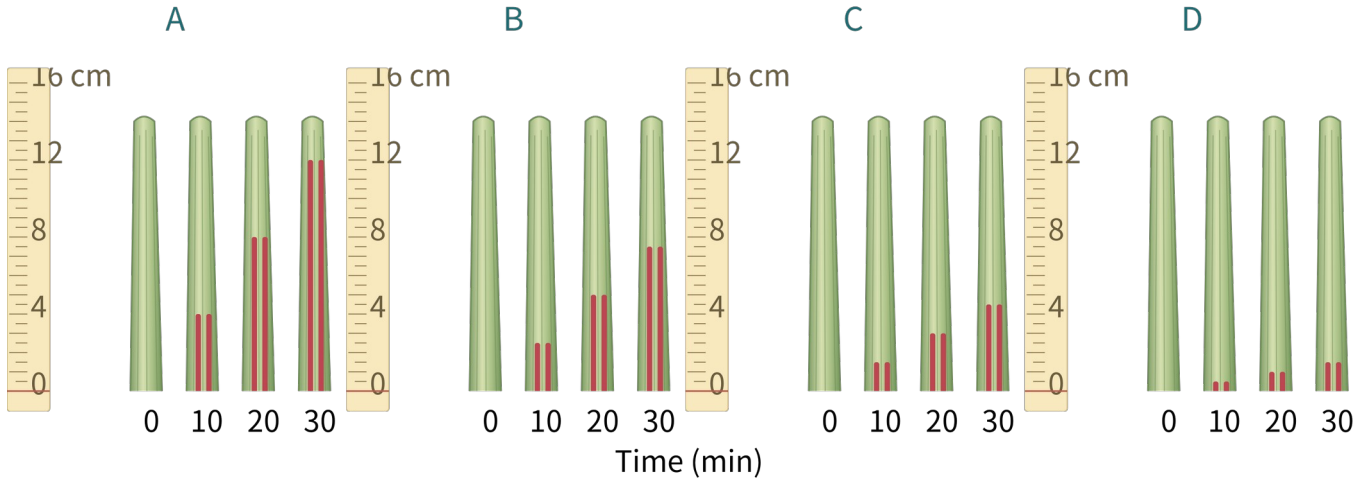
(Hint: how is the control setup different from the experiment setup?)

Reference answer: Compare leafy B with leafless D at the same 20 cm lamp distance, keeping all other conditions equal.

Part 2 | Results and analysis

4. Your measurement record

The diagrams show water uptake in the plant xylem of four setups at different times. Read the height of the top of each red water trace at each time, to one decimal place (cm).



Time (min)	A	B	C	D
0	**0.0**	**0.0**	**0.0**	**0.0**
10	**4.0**	**2.5**	**1.5**	**0.5**
20	**8.0**	**5.0**	**3.0**	**1.0**
30	**12.0**	**7.5**	**4.5**	**1.5**

5. Calculate the average upward speed of the top of the red water trace

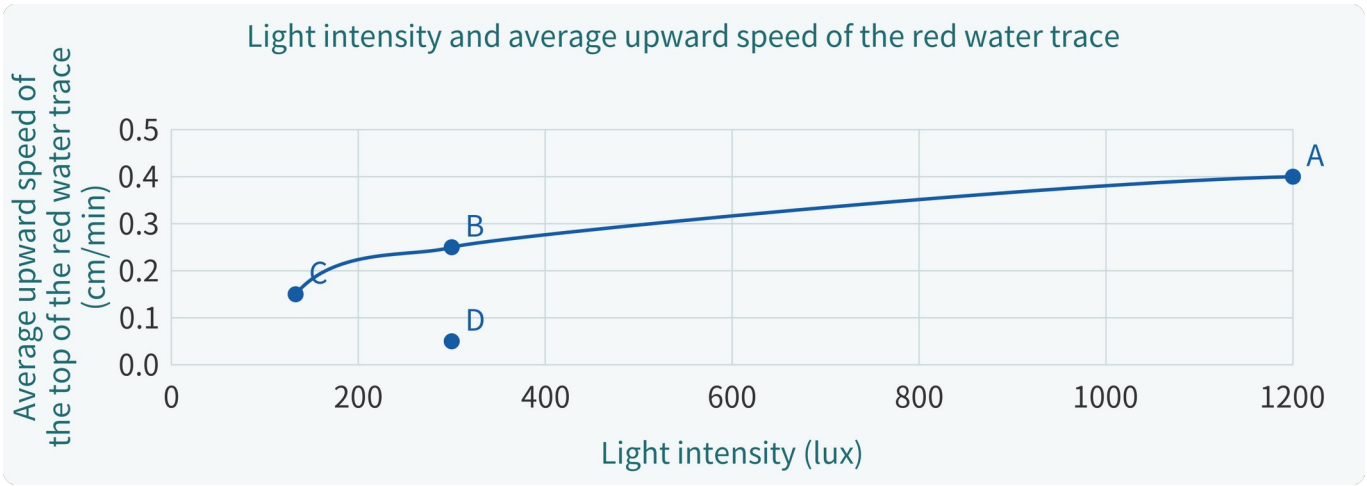
(30 min reading − 0 min reading) ÷ 30; give two decimal places (cm/min).

A	B	C	D
0.40	**0.25**	**0.15**	**0.05**

6. Light intensity and water-trace speed

Plot A–D from your calculations. Join A–C with a smooth trend line; keep D separate.

A: 1200 lux B: 300 lux C: 133 lux D: 300 lux



Part 2 | Results and analysis (continued)

7. How is light intensity (光強度) related to the upward speed of the water trace?

As light intensity (光強度) increases, the average upward speed of the red water trace usually will （**increase** / decrease / show no clear change / show no consistent trend）.

8. What is the role of leaves?

At the same 20 cm lamp distance, the red-trace speed in leafy setup B, compared with leafless setup D, is （**faster** / slower / similar）.

9. What is the role of leaves?

This comparison supports the idea that leaves （**help water move upwards** / stop water moving upwards / have no effect on water moving upwards）.

10. What can this experiment tell us?

We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?

☑ **The red water trace indicates water uptake and upward water transport, but does not represent leaf transpiration rate (蒸騰速率) (the rate of water loss through stomata (氣孔)).**

☐ The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率).

☐ If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).

Part 3 | Integrated explanation

11. Using the experimental results and learning points, explain how light intensity (光強度) and leaves affect water transport, and explain how transpiration (蒸騰) and transpiration pull (蒸騰拉力) arise.

Reference answer: Other conditions equal, stronger light usually opens stomata, increasing transpiration and water transport. Water evaporates from mesophyll water films into air spaces; vapour diffuses down its concentration gradient through stomata: transpiration. Loss lowers mesophyll water potential; xylem replenishes water. Removal creates transpiration pull, drawing water up xylem vessels. Leaves provide the main transpiring surface, so leafy B has a faster trace than leafless D at equal distance.